

PERTURBATIVE SOLUTIONS FOR NON-LINEAR FIXED-BED ADSORPTION—I. APPROXIMATE ANALYTICAL SOLUTIONS FOR ASYMPTOTIC FRONTS

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Abstract The authors present a new theoretical approach for a general model of non-linear chromatography in the presence of axial dispersion and mass transfer resistances, based on perturbation analysis. They first derive approximate analytical expressions for steady (constant-pattern) fronts, which allow them to clarify the empirical idea that the HTUs (height of transfer units) for mass transfer and axial dispersion are additive. This result is in fact an approximation for which they derive an applicability criterion. They also show how the concentration of the stoichiometric point of the constant pattern front depends on the kinetic mechanism. Computation of higher-order terms allows them to derive an approximation to the transient front, which, by comparison with numerical and experimental results, has been found to be very accurate.

INTRODUCTION

The design of industrial adsorption and chromatography operations requires a good knowledge of column dynamics; in particular, it is essential to analyse the influence of experimental conditions (geometry, fluid velocity, concentrations . . .) and model parameters (thermodynamics, hydrodynamics, mass transfer rates . . .) on the breakthrough curve. The material balance equations of non-linear chromatography in the presence of mass transfer resistances and axial hydrodynamic dispersion exhibit a non-trivial mathematical structure, and are not in general solvable analytically in terms of elementary functions. In some special cases (Vermeulen and Hiester, 1952) solutions exist in terms of Bessel functions, but these expressions are rather inconvenient and do not yield simple information. Direct numerical integration of the set of partial differential equations is in general possible (Costa *et al.*, 1986) but may be expensive in computing time. In any case, it is generally difficult to determine general trends from isolated numerical simulations.

In what follows, we will adopt an approach which can be considered as intermediate between numerical and exact analytical integration. The methods we use here have already proved very powerful in quantum (Landau and Lifschitz, 1969a), statistical (Landau and Lifschitz, 1969b) and non-linear physics (Baker and Graves-Morris, 1981); they are based on perturbation analysis (Bender and Orszag, 1978) and essentially consist in determining an approximate analytical solution by splitting the original problem into a series of simpler ones, each of these being solvable in terms of elementary functions.

We shall be concerned first with the derivation of approximate expressions for the shock layer in non-equilibrium systems with axial dispersion and a rather

general mass transfer law, and then we shall investigate their relevance by comparing the results with those provided by direct numerical integration. This analytical description will then allow us to derive some general results concerning the stoichiometric point of the breakthrough curves and the transient behaviour of the fronts.

BASIC FORMULATION AND DEFINITIONS

We consider a fixed bed initially uniformly pre-saturated. Let C_i be the sorbate concentration in the interstitial fluid, and q_i the concentration on the solid, in equilibrium with C_i :

$$q_i = f_{eq}(C_i). \quad (1)$$

The isotherm $f_{eq}(C)$ is assumed to be "favourable", that is type I in the BET classification:

$$f''_{eq}(C) < 0.$$

At $t = 0$, a positive step input of concentration $C_e > C_i$ is applied to the column entry point. Under these conditions, it has been shown (Rhee and Amundson, 1972) that the front asymptotically approaches, as the column length goes to infinity, a limit or constant pattern which we call the stationary front, and which propagates in the column without deformation. We define normalized concentrations X in the fluid and Y in the solid as

$$X \equiv \frac{C - C_i}{C_e - C_i} \quad (2)$$

$$Y \equiv \frac{q - q_i}{q_e - q_i} \quad (3)$$

$$q_e = f_{eq}(C_e) \quad (4)$$

and the non-dimensional isotherm $f(X)$ as

$$f(X) = \frac{f_{eq}(C) - q_i}{q_e - q_i} = Y_{eq}. \quad (5)$$

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Mass balance equations

If the hydrodynamic dispersion is characterized by a Peclet number, $Pe = uL/D$, the non-dimensional mass balance equation may be written in the form

$$\left(\frac{\partial X}{\partial Z}\right)_T + \left(\frac{\partial Y}{\partial T}\right)_Z = \frac{1}{\lambda Pe} [\partial_T - \lambda \partial_Z]^{(2)} X$$

$$= \frac{1}{\lambda Pe} \left[\frac{\partial^2}{\partial T^2} - 2\lambda \frac{\partial^2}{\partial Z \partial T} + \lambda^2 \frac{\partial^2}{\partial Z^2} \right] X \quad (6)$$

where λ is the capacity ratio

$$\lambda = \frac{1 - \varepsilon}{\varepsilon} \frac{q_e - q_i}{C_e - C_i}$$

The brackets in the right-hand sides of eq. (6) designate a differential operator which applies here to the variable X . In the middle expression, the (2) corresponds to the formal square of the operator, as expressed explicitly in the right-hand side. Z and T are non-dimensional length and time variables (see Notation). ∂_T stands for $\partial/\partial T$, and similarly for ∂_Z .

The adsorption kinetics are assumed to be of the general type

$$\left(\frac{\partial Y}{\partial T}\right)_Z = F(X, Y). \quad (7)$$

The boundary conditions are, in (z, t) variables:

$$t = 0, \quad Y = 0, \quad 0 \leq z \leq 1 \quad (8a)$$

$$z = 0, \quad 1 = X - \frac{1}{Pe} \frac{\partial X}{\partial z} \Big|_t \quad (8b)$$

$$z = 1, \quad \frac{1}{Pe} \frac{\partial X}{\partial z} \Big|_t = 0 \quad (8c)$$

and, in (Z, T) variables:

$$T = \frac{Z}{\lambda}, \quad 0 \leq Z \leq \lambda, \quad Y = 0 \quad (8d)$$

$$T \leq 0, \quad Z = 0, \quad 1 = X - \frac{1}{Pe} \left(\lambda \frac{\partial X}{\partial Z} - \frac{\partial X}{\partial T} \right) \quad (8e)$$

$$T > -1, \quad Z = \lambda, \quad \frac{1}{Pe} \left(\lambda \frac{\partial X}{\partial Z} - \frac{\partial X}{\partial T} \right) = 0. \quad (8f)$$

Change of representation

It has been shown (Rhee and Amundson, 1972) that the solutions $X(Z, T)$ and $Y(Z, T)$ are monotonic functions of both Z and T . This enables one to consider the equivalent representation

$$Z = Z(X, T) \quad (9a)$$

$$Y = Y(X, T). \quad (9b)$$

When these expressions are known, they give the "abscissa" Z of any given concentration X at any time T , and thus allow a straightforward construction of the front. In what follows, we are thus essentially concerned with obtaining suitable expressions for Z .

With the change of functions so defined, eqs (6) and (7) become

$$\frac{1}{Z_X} + F(X, Y) = \frac{1}{\lambda Pe} [\lambda \partial_T - \lambda \partial_Z]^{(2)} X \quad (10)$$

$$Y_T - Y_X \frac{Z_T}{Z_X} = F(X, Y). \quad (11)$$

Z_X , for example, stands for $\left(\frac{\partial Z}{\partial X}\right)_T$, and similarly for Z_T , Y_X and Y_T .

Perturbation analysis

In the limit $T = \infty$, it has been shown that the front asymptotically approaches a stable pattern (Rhee and Amundson, 1972) described by an equation of the form

$$Z = T + \Phi(X). \quad (12)$$

The function $\Phi(X)$ has been determined in the plug-flow case $Pe = \infty$ (Lapidus and Rosen, 1954).

For finite times, the perturbative approach consists in expanding $Z(T, X) - T - \Phi(X)$ in a series of separable functions

$$Z(T, X) - T - \Phi(X) = \sum_{n=1}^{\infty} A_n(T) Z^{(n)}(X) \quad (13)$$

with the condition $A_{n+1} \ll A_n$, i.e.

$$\lim_{T \rightarrow \infty} \frac{A_{n+1}(T)}{A_n(T)} = 0. \quad (14)$$

Asymptotic matching techniques (Bender and Orszag, 1978) show that a suitable form for $A_n(T)$ is T^{-n} and therefore we write

$$Z(T, X) = T + \sum_{n=0}^{\infty} \frac{Z^{(n)}(X)}{T^n} \quad (15)$$

with

$$Z^{(0)}(X) = \Phi(X). \quad (16)$$

Similarly, we expand the variable Y :

$$Y(T, X) = Y^{(0)}(X) + \frac{Y^{(1)}(X)}{T} + \dots \quad (17)$$

The part $T + Z^{(0)}(X)$ represents the stationary front and the first term of the expansion $Z^{(1)}(X)/T$ represents the leading behaviour of the transient front.

In what follows, we will derive approximate analytical expressions for $Z^{(0)}(X)$, $Z^{(1)}(X)$, ... in the case of axial hydrodynamic dispersion and kinetics of the type defined by eq. (7).

LEADING ORDER FRONT: THE CONSTANT PATTERN

Derivation of the constant-pattern equation

Substituting expansions (15)–(17) into eqs (10) and (11) and neglecting terms of order at least $1/T$ leads

after some manipulations to

$$\frac{1}{Z^{(0)}} + F[X, Y^{(0)}(X)] = \frac{1}{\lambda Pe} [\lambda \partial_Z - \partial_T]^{(2)} X \quad (18)$$

$$Y = X - \frac{(\lambda + 1)^2}{Pe \lambda} \frac{1}{Z^{(0)}} \quad (19)$$

This system is not solvable in general, even for simple kinetic functions F (Rhee and Amundson, 1972). Here again, we will adopt the perturbative point of view: the constant pattern is attained for very long columns, and so the above equations are valid for $Pe \gg 1$.

This remark leads us to expand

$$F\left[X, X - \frac{(\lambda + 1)}{Pe \lambda} \frac{1}{Z^{(0)}}\right] \text{ in powers of } \frac{1}{Pe}.$$

The second-order Taylor expansion in powers of $\frac{1}{Pe}$ is

$$Z^{(0)}(X) = T - \int \frac{dX}{F(X, X)} + \frac{(\lambda + 1)^2}{\lambda Pe} \int \frac{F_Y(X, X) dX}{F(X, X)} + \frac{(\lambda + 1)^2}{\lambda Pe} \int \frac{U^{(0)}(X) dX}{F(X, X)} + C^{(e)} + O\left(\frac{1}{Pe^2}\right) \quad (28)$$

written as

$$F\left[X, X - \frac{\lambda + 1}{\lambda Pe} \frac{1}{Z^{(0)}}\right] = F(X, X) - \frac{(\lambda + 1)^2}{\lambda Pe} \frac{1}{Z^{(0)}} F_Y(X, X) + \frac{1}{2} \frac{(\lambda + 1)^4}{\lambda^2 Pe^2} \frac{1}{Z^{(0)^2}} F_{YY}(X, X) + O\left(\frac{1}{Pe^3}\right) \quad (20)$$

The notation $O\left(\frac{1}{Pe^3}\right)$ represents all terms of order $\frac{1}{Pe^3}$ at least.

Putting $\frac{1}{Z^{(0)}} = U^{(0)}(X)$, eq. (18) becomes at order

$$\frac{1}{Pe} \frac{(\lambda + 1)^2}{\lambda Pe} U^{(0)} U'^{(0)} + U^{(0)} \left[\frac{(\lambda + 1)^2}{\lambda Pe} F_Y(X, X) - 1 \right] = F(X, X) + O\left(\frac{1}{Pe^2}\right) \quad (21)$$

The perturbative solution $U^{(0)}(X)$ of this equation takes the form

$$U^{(0)}(X) = U^{00}(X) + \frac{(\lambda + 1)^2}{\lambda Pe} U^{01}(X) + O\left(\frac{1}{Pe^2}\right) \quad (22)$$

with

$$U^{00}(X) = \frac{F(X, X)}{\frac{(\lambda + 1)^2}{\lambda Pe} F_Y(X, X) - 1} \quad (23)$$

$$U^{01}(X) = - \frac{[U^{00}(X)]^2 U'^{(0)}(X)}{F(X, X)} \quad (24)$$

Recalling that $U^{(0)}(X) = \frac{1}{Z^{(0)}}$ we can write

$$Z^{(0)}(X) = Z^{00}(X) + \frac{(\lambda + 1)^2}{\lambda Pe} Z^{01}(X) + O\left(\frac{1}{Pe^2}\right) \quad (25)$$

with

$$Z^{00}(X) = \int \frac{dX}{U^{00}(X)} \quad (26)$$

and

$$Z^{01}(X) = \int \frac{U'^{(0)}(X) dX}{F(X, X)} \quad (27)$$

The above equations finally lead to the following development which represents the constant-pattern solution:

The integration constant is determined by the total mass balance equation (Klein, 1985):

$$\int_0^1 [Z^{(0)}(X) - T] dX = 0. \quad (29)$$

Equations (28) and (29) fully determine the front. Notice that we have not as yet assumed any explicit form for the equilibrium isotherm and the transfer kinetics. We shall now seek more explicit expressions.

Applications

The linear driving force models (Glueckauf and Coates 1947).

External resistance

For a pure external mass transfer resistance, $F(X, Y)$ takes the form

$$F(X, Y) = \frac{\xi St_e}{\lambda} [X - g(Y)] \quad (30)$$

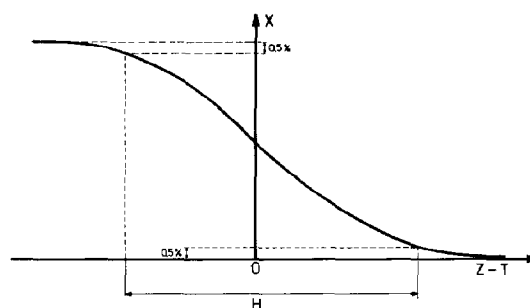


Fig. 1. Definition of the non-dimensional height of the transfer zone, H .

where $St_e = \frac{k_e L}{u}$ is a Stanton number, and $g(Y) = f^{-1}(Y)$ the reciprocal of the isotherm function.

Substitution into eq. (28) gives

$$Z^{(0)}(X) = T + \left[\frac{\lambda}{St_e} + \frac{(\lambda + 1)^2}{\lambda Pe} \right] \int \frac{dX}{g(X) - X} + \frac{(\lambda + 1)^2}{\lambda Pe} \ln |g(X) - X| + O \left[\frac{(\lambda + 1)^2}{Pe St_e} \right]. \quad (31)$$

By analogy with distillation one often defines a non-dimensional "height of transfer zone" (HTZ) by

$$HTZ = H = \int_{0.005}^{0.995} |Z^{(0)} - T| dX.$$

In the following, we shall simply designate the HTZ by H .

Figure 1 illustrates the graphical meaning of this quantity on the non-dimensional representation of the front. Notice that the variables Z and T used here are such that the front is centered at $Z = T$.

Using eq. (31), one finds that

$$H = \overbrace{\left[\frac{\lambda}{St_e} + \frac{(\lambda + 1)^2}{\lambda Pe} \right]}^{HTU} \overbrace{\int_{0.005}^{0.995} \frac{dX}{X - g(X)}}^{NTU} + \frac{(\lambda + 1)^2}{\lambda Pe} \ln \left| \frac{g(0.995) - 0.995}{g(0.005) - 0.005} \right| + O \left[\frac{(\lambda + 1)^2}{Pe St} \right]. \quad (32)$$

In the first term, the integral may be considered as the number of transfer units (NTUs) as defined in the usual way, and the bracket as the height of transfer unit (HTU). Thus it appears to represent the addition of the HTUs corresponding to mass transfer (λ/St_e) and axial dispersion. The second term is of order $1/Pe$ and equivalent to

$$\frac{(\lambda + 1)^2}{\lambda Pe} \ln \left| \frac{1 - g'(1)}{1 - g'(0)} \right|$$

which may not be negligible: in other words, perturbation theory provides the leading order correction to HTU additivity, and so provides a useful additivity criterion. We shall next illustrate more quantitatively this effect on the case of internal resistance.

Internal resistance

In the case of the intraparticle linear driving force, F is

$$F(X, Y) = St_i [f(X) - Y] \quad (33)$$

where St_i is a Stanton number. One can notice here that $F_{YY} = 0$ and so eq. (21) is valid through all orders in $1/Pe$.

Letting $\delta = \frac{(\lambda + 1)^2}{Pe \lambda}$, the solution becomes

$$Z^{(0)}(X) = T + \left[\frac{(\lambda + 1)^2}{Pe \lambda} + \frac{1}{St_i} \right] \int \frac{dX}{X - f(X)}$$

$$+ \frac{\delta(1 - \delta)}{1 + \delta St_i} \ln |f(X) - X| - \frac{\delta^2 St_i}{(1 + \delta St_i)^3} [f(X) - X]$$

$$- \frac{2\delta^2 St_i}{(1 + \delta St_i)^3} \int \frac{[1 - f'(X)]^2}{X - f(X)} dX + O \left(\frac{1}{Pe^3} \right). \quad (34)$$

In general, the last two terms are negligible as will be shown in the following. The same conclusions as above hold in this case: additivity of HTUs is justified only when all terms after the first are negligible.

Additivity of HTUs had been conjectured after numerical integration (Rhee and Amundson, 1972). These authors considered a Langmuir isotherm:

$$f(X) = \frac{(1 + R)X}{1 + RX} \quad (35)$$

with $R = 1/44 \cong 0.22727$. They used $\lambda = 5.0505$, which is a rather small value. For Pe and St_i ranging from 20 to 160, they found that the non-dimensional transfer zone height H is given by the first term:

$$H \cong \left[\frac{(\lambda + 1)^2}{Pe} + \frac{1}{St_i} \right] \int_{0.005}^{0.995} \frac{dX}{f(X) - X} \quad (36)$$

with an accuracy which was always better than 0.13%.

Using the next terms, we found an agreement with the last digit given by these authors. The isotherm Rhee and Amundson used is practically linear in the range considered ($R = 0.23$). For an isotherm with larger curvature, e.g. $R = 2$, the ratio ρ of the second term to the first in the development of H is

$$\rho = 0.13 \left| \frac{St_i \delta (1 - \delta)}{(1 + \delta St_i)^2} \right|. \quad (37)$$

In many experimental systems, λ is of the order of 500 or more (Vermeulen and Hiester, 1952); with $Pe = 200$, this implies $\delta \cong 2.5$. Thus

$$\rho \cong \frac{0.5 St_i}{(1 + 2.5 St_i)^2}. \quad (38)$$

With $St_i = 0.4$, ρ has a maximum of 10%, which represents the error made when neglecting the terms after the first in the development of eq. (34). For a smaller Pe , say $Pe = 20$, $\delta = 25$, so that

$$\rho = \frac{78 St_i}{(1 + 25 St_i)^2} \quad (39)$$

which is maximum for $St = 0.04$ where $\rho = 78\%$.

The latter case can be considered as a limit: the calculation is valid for deep beds, and $Pe = 20$ means a large hydrodynamic dispersion, and $St = 0.04$ means very slow kinetics. Anyway, such a system may exist and, in this case, the additivity suggested in (Rhee and Amundson, 1972) no longer holds.

Consequences

Perturbative analysis allowed us to derive analytically and thus generally the result found on numerical examples by Rhee and Amundson and furthermore to give an analytical criterion for its application. The above results as well as numerous comparisons with numerical simulations (Gorius and Bailly, 1987) show that the stationary front is in general given with a good approximation by

$$Z^{(0)}(X) \cong T + \left[\frac{(\lambda + 1)^2}{\lambda Pe} + \frac{\alpha}{St} \right] \int_{X_s}^X \frac{dX}{\varphi(X)} \quad (40)$$

where $\varphi(X)$ is the driving force, and α a constant; we have, for example

for external resistance:

$$\alpha = \frac{\lambda \varepsilon}{1 - \varepsilon} = \lambda / \xi \quad \text{and} \quad \varphi(X) = g(X) - X \\ = f^{-1}(X) - X \quad (41a)$$

for internal resistance:

$$\alpha = 1 \quad \text{and} \quad \varphi(X) = X - f(X). \quad (41b)$$

The constant X_s is called the stoichiometric concentration of the stationary front (Klein, 1985), and is determined by

$$\int_0^1 [Z^{(0)}(X) - T] dX = 0. \quad (42)$$

Putting $\int \frac{dX}{\varphi(X)} = \Phi(X)$, this is written as

$$\Phi(X_s) = \int_0^1 \Phi(X) dX. \quad (43)$$

For a given linear driving force F , this is the equation giving the concentration X_s of the stoichiometric point of the stationary front.

Stoichiometric concentration. In what follows, we will depict some interesting results concerning the stoichiometric concentration X_s in the case of linear internal and external driving forces, first on the example of a Langmuir isotherm, and then in the general case.

Langmuir isotherm

Let

$$f(X) = \frac{(R + 1)X}{1 + RX}. \quad (44)$$

For an internal linear driving force, $\varphi(X) = X - f(X)$, eq. (43) yields

$$\frac{(1 - X_s)^{1+R}}{X_s} = \exp(-R). \quad (45)$$

From this formula, it is easy to check that

— For a linear isotherm ($R = 0$), the stoichiometric point is at half-height: $X_s = 0.5$.

— For a “rectangular” or “irreversible” isotherm ($R = \infty$), $X_s = 1 - \exp(-1) = 0.632$.

— For any finite, non-zero value of R , X_s is intermediate between the above values, thus

$$0.5 \leq X_s \leq 0.632.$$

For an external driving force, eq. (42) is written as

$$\frac{(1 - X_s)^{\frac{1}{1+R}}}{X_s} = \exp\left(\frac{R}{1+R}\right). \quad (46)$$

The substitution $X_s \rightarrow 1 - X_s$ transforms this equation into eq. (45), which shows that, for the same value of R , the value of X_s is in the internal case the mirror image of the value obtained in the external case with respect to the limit value $X_s = 0.5$, as shown on Fig. 2(a). Thus

$$0.368 \leq X_s \leq 0.5$$

Generalisation

This symmetry is a particular feature due to the symmetry of Langmuir isotherms, which no longer holds for other ones. For example, Fig. 2(b) shows X_s vs α for a Freundlich isotherm $f(X) = X^2$. However, we have been able to show the following general results (Gorius, 1988).

Consider a family of sufficiently regular favourable isotherms $f(a, X)$ depending on a parameter “ a ” and including as limit functions a linear isotherm [$f(a_0, X) = X$] and an irreversible one [$f(a_1, X) = 1, X \neq 0$].

Let $X_s(a)$ be the stoichiometric concentration of the stationary front obtained for the isotherm $f(a, X)$. Then

— for an internal linear driving force

$$X_s(a_0) \leq X_s(a) \leq X_s(a_1) = 1 - \exp(-1)$$

$$\frac{dX_s}{da} \text{ has the sign of } \frac{\partial f}{\partial a}$$

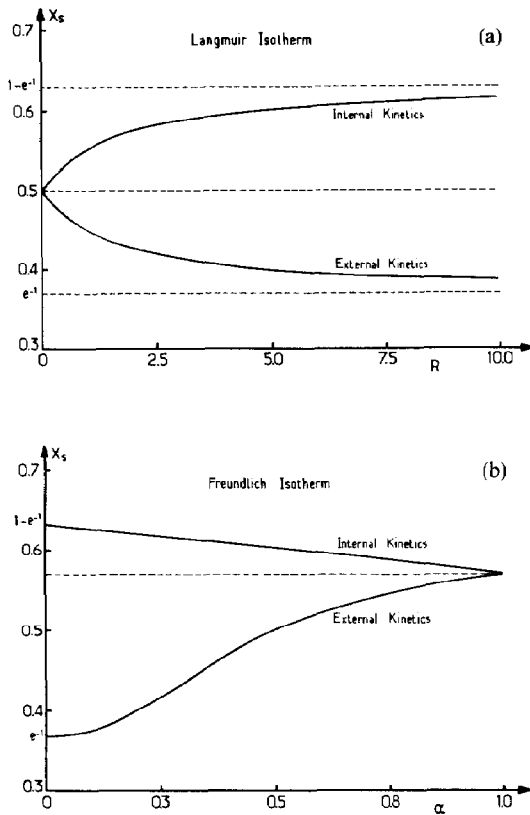


Fig. 2. (a) Stoichiometric point X_s of the stationary front as a function of R for a Langmuir isotherm $f(X) = (1 + R)X/(1 + RX)$. (b) Stoichiometric point of the stationary front as a function of α for a Freundlich isotherm $f(X) = X^\alpha$.

— for an external linear driving force

$$X_s(a_1) = \exp(-1) \leq X_s(a) \leq X_s(a_0)$$

$$\frac{dX_s}{da} \text{ has the sign of } -\frac{\partial f}{\partial a}.$$

As a consequence, the stoichiometric concentration for an external (internal) linear driving force is always below (above) its limit value obtained for a linear isotherm, and always above (below) its limit value obtained for an irreversible equilibrium.

This result can be used as a criterion for evaluating the governing type of kinetic limitation, since for a given isotherm $f(a, X)$, $X_s(a_0)$ and $X_s(a_1)$ are easily calculated in each kinetic limitation case.

TRANSIENT BEHAVIOUR

The expansion in powers of T^{-1} [eq. (15)] allows direct calculation of the first transient terms, once the stationary front $Z^{(0)}(X)$ is known.

The calculation consists in inserting the expansions (15)–(17) into eqs (10) and (11) neglecting at order $O(T^{-n})$ all terms of order at least $O(T^{-(n+1)})$.

For example, at order $O(T^{-1})$, the equations in $Z^{(1)}$ and $Y^{(1)}$ are

$$-\frac{Z_X^{(1)}}{[Z_X^{(0)}]^2} + F_Y[X, Y^{(0)}(X)] = O\left(\frac{1}{Pe}\right) \quad (47)$$

$$-Y_X^{(0)} \frac{Z_X^{(1)}}{[Z_X^{(0)}]^2} - \frac{Y_X^{(1)}}{Z_X^{(0)}} = F_Y[X, Y^{(0)}(X)]. \quad (48)$$

We expand these equations in power of $1/Pe$:

$$Z^{(1)}(X) = \sum_{n=0}^{\infty} \frac{1}{Pe^n} Z^{1n}(X) \quad (49)$$

$$Y^{(1)}(X) = \sum_{n=0}^{\infty} \frac{1}{Pe^n} Y^{1n}(X). \quad (50)$$

As before the asymptotic matching technique leads to the leading order solution:

$$Z^{(10)} = \int (Z_X^{(0)})^2 dX. \quad (51)$$

Taking account of the next terms Z^{11} and Z^{12} shows that the transient front is given by

$$Z(X, T) = Z^{(0)}(X) + \frac{1}{T} \int [Z^{(0)}(X)]^2 dX + O\left(\frac{1}{T^2}, \frac{1}{Pe^3}\right) \quad (52)$$

where $Z^{(0)}(X)$ is given by eq. (28).

This equation gives the leading order correction to the constant-pattern solution taking into account transient effects.

For example, for an internal driving force $F = St_i[f(X) - Y]$, the results of Part II of the present work show that the front is given with a good approximation by

$$Z(X, T) \cong T + \left[\frac{1}{St_i} + \frac{(\lambda + 1)^2}{Pe\lambda} \right] \int_{X_s}^X \frac{dX}{X - f(X)} + \frac{1}{T} \left[\frac{1}{St_i} + \frac{(\lambda + 1)^2}{Pe\lambda} \right]^2 \int_{X_1}^X \frac{dX}{[X - f(X)]^2}. \quad (53)$$

In eq. (53), X_s is evaluated from the steady-state solution, using eq. (43) as indicated earlier. X_1 in the second integral is an integration limit determined so as to satisfy the overall material balance in the transient state as well. This is done using again eq. (43), but with $\varphi(X)$ defined as the denominator of the second integral, i.e. $\varphi(X) = [X - f(X)]^2$.

This result is supported by numerous comparisons with numerical integrations. For example, Table 1 shows the maximum relative excursion between the numerical solution and that given by eq. (52). These

Table 1.

Pe	St	$E_1^†$	$E_2^‡$
120	0.01	2.9×10^{-3}	2.5×10^{-4}
120	0.1	1.7×10^{-3}	5.2×10^{-5}
120	1.0	4.4×10^{-4}	2.4×10^{-5}
120	10.0	1.3×10^{-4}	1.7×10^{-6}
120	100.0	5.5×10^{-5}	5.5×10^{-6}
10	0.01	3.1×10^{-3}	7.0×10^{-4}
10	1.0	9.1×10^{-4}	5.5×10^{-5}
10	100.0	7.2×10^{-5}	4.0×10^{-6}
1	1.0	3.4×10^{-3}	2.9×10^{-4}

†E_1 is the relative discrepancy between numerical result and order:

$$O\left(\frac{1}{T}, \frac{1}{Pe^3}\right).$$

‡E_2 is the same for $O\left(\frac{1}{T^2}, \frac{1}{Pe^3}\right).$

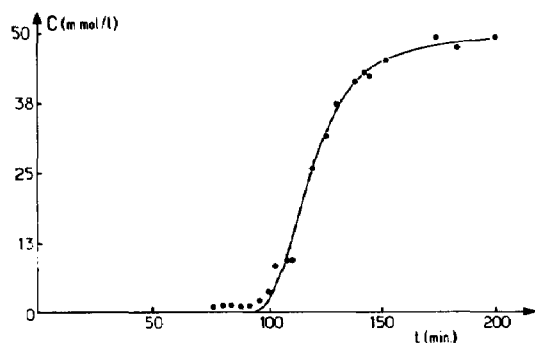


Fig. 3. Comparison between theory [eq. (53)] and experiment: adsorption of phenylacetic acid on Duolite ES 861. We used $Pe = 120$ and $St = 0.8$ for the theoretical curve (continuous line).

simulations were made for a Freundlich isotherm $f(X) = X^{0.54}$ with $\lambda = 50$.

The result is always very satisfactory; moreover computation of the next term of order

$$O\left(\frac{1}{T^2}, \frac{1}{Pe^2}\right)$$

in eq. (52) shows that the departure from the numerical result (E_2 in Table 1) is always much lower than this term. In other words, series (15) converges very rapidly.

We have tried to apply formula (53) to our experimental system: adsorption of phenylacetic acid on Duolite ES 861. The column used has an internal diameter of 1.4 cm and a height of 33 cm. The particle diameter was centered around 0.06 cm. We used $C_i = 0$, $C_e = 50$ mmol/l, and measured $q_e = 1.21$ mol/l, which leads to $\lambda = 36.4$ with $\varepsilon = 0.4$. The adsorption isotherm is well described by a Freundlich equation with exponent 0.54 as above. The driving force was taken as an internal linear one.

Axial dispersion was estimated by correlations

(Wen and Fan, 1975) giving $Pe \cong 120$ and the mass transfer coefficient k_i was determined by the shallow bed technique giving $St = 0.8$. Figure 3 shows the experimental points and the curve calculated by eq. (53). Agreement is very satisfactory, as in every other experimental run. Let us emphasize the following points:

— no adjustment of parameters was necessary;

— equation (53) can be used on a pocket calculator, whereas direct numerical integration of the mass balance equations takes several hours on a Honeywell-Bull Mini 6-92 mini computer for the same result.

CONCLUSIONS

The present work has shown how advanced mathematical techniques (asymptotic and perturbative analysis) allow one to write accurate approximate analytical solutions to the complex problem of non-linear chromatography with non-ideal hydrodynamics and non-trivial kinetics.

We have derived expressions for the stationary front, which allowed us to investigate the applicability domain of the additivity of HTUs corresponding to axial dispersion and mass transfer. We also showed how the value of the stoichiometric concentration of the stationary front allows a discrimination between pure internal or external linear driving forces.

The present analysis was made for long columns, and for favourable isotherms. The mathematics of the unfavourable case are much simpler (Gorius, 1988). In subsequent papers, we will show how it is possible to include the present theory in a more general one, valid for columns of arbitrary length and arbitrary isotherms.

Perturbative analysis of the front's transient behaviour provided simple analytical expressions, the results of which agreed satisfactorily with both numerical and experimental runs.

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NOTATION

C	concentration in the fluid phase, mol/l
C_e	inlet concentration, mol/l (constant)
C_i	initial concentration, mol/l (constant)
D	axial dispersion coefficient, $m^2 s^{-1}$
f	non-dimensional isotherm
	$f(X) = \frac{f_{eq}(C) - q_i}{q_e - q_i}$
f_{eq}	isotherm, mol/l
F	non-dimensional mass transfer driving force
H	non-dimensional height of transfer zone (defined in Fig. 1)
k_e	external mass transfer coefficient, s^{-1}
k_i	internal mass transfer coefficient, s^{-1}
L	bed length, m
Pe	Peclet number (uL/D)

q	concentration in the solid phase, mol/l
q_e	$f_{eq}(C_e)$
q_i	$f_{eq}(C_i)$
R	non-dimensional Langmuir constant
St_e	external Stanton number ($k_e L/u$)
St_i	internal Stanton number ($k_i L/u$)
t	non-dimensional time measured in L/u units
t'	usual time, s [$t' = (L/u)t$]
T	$t - z$
u	interstitial fluid velocity, m/s
V_s	stationary front non-dimensional velocity [$1/(1 + \lambda)$]
X	dimensionless normalized fluid concentration $\left(\frac{C - C_i}{C_e - C_i} \right)$
X_S	stoichiometric concentration of the stationary front
Y	dimensionless normalized solid phase concentration $\left(\frac{q - q_i}{q_e - q_i} \right)$
z	non-dimensional abscissa (in L units)
z'	abscissa from bed inlet (Lz)
Z	λz

Greek letters

α	Freundlich exponent
δ	$(\lambda + 1)^2 / \lambda Pe$
ϵ	bed porosity
λ	$\frac{1 - \epsilon}{\epsilon} \frac{q_e - q_i}{C_e - C_i}$
ξ	$\frac{1 - \epsilon}{\epsilon}$

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